

FERN & STEAM

Courtyard Edition

352 pieces / 24 x 24 studs / arched conservatory cafe



Actual LDraw geometry render

A translation of the approved design concept using exact elements listed in the donor collection. This is a digital construction draft for review and a trial build, not physically tested instructions. The editable LDraw file is the authoritative placement reference.

Before you build

What has been checked

The combined bill of materials fits the conservative donor allowance. Ordinary brick, plate and tile bodies have no nominal box overlaps. The stud attachment graph is connected, and compatible window inserts use the documented frame transforms. Actual mesh bounds have been checked against the authored placements.

What remains to be checked

A physical trial build, clutch strength, load stability, insertion access, and a complete solid collision check remain outstanding. The targeted accessory mesh probe is additional evidence, not a general collision certification. Studio may prompt you to update newer parts.

Inventory assumption

Each exact element ID counts once per owned donor-set copy. This is a conservative allowance from set membership, assuming complete accessible sets and no other reservations. Actual set quantities may be higher. The donor list is a reproducible allocation under this rule, not a requirement to dismantle that many sets.

How to read the coordinates

Face the cafe from the bench side. The rear-left base corner is (0,0). X runs left to right and Y runs rear to front, both in studs; the base occupies $X=0..24$ and $Y=0..24$. Z is upward in plate heights, measured at the bottom of the part body. One brick is 3 plates. For ordinary parts X/Y marks the lower-left footprint corner, not its center.

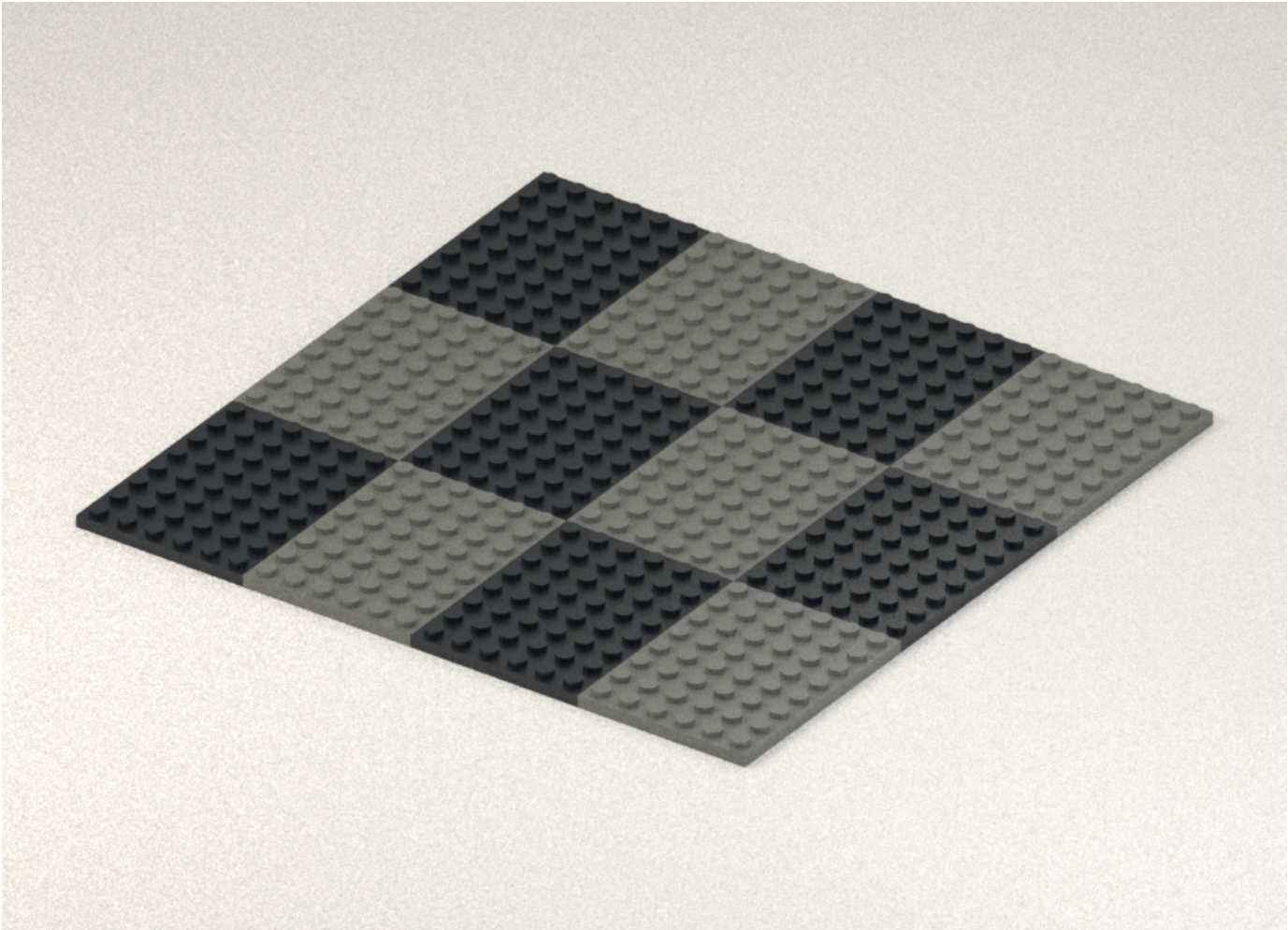
Yaw is the absolute LDraw rotation about its downward Y axis, viewed from above. The footprint column gives the final X by Y size, making rectangular-part orientation unambiguous. Round-mounted plants may use 45-degree yaw increments. For panes, attach to the named parent frame with the listed local LDraw offset instead of using footprint coordinates. 20 LDraw units = 1 stud; 8 units = 1 plate.

Assembly notes

Build the counter before fitting the arches and roof. Insert panes into their frames before the cornice. Assemble the lantern support rails and eaves as a small subassembly, then lower it onto the cornice; do not expect every loose rail to stand on its own. Hold the model by the base. Inspect the first trial assembly before applying force to any uncertain connection.

01 / Foundation

Arrange the twelve 6 × 8 plates on a flat surface.



Add 12 pieces in this stage

Element	Color / geometry	Qty
303626	Black / 3036.dat	6
4210794	Dark Bluish Gray / 3036.dat	6

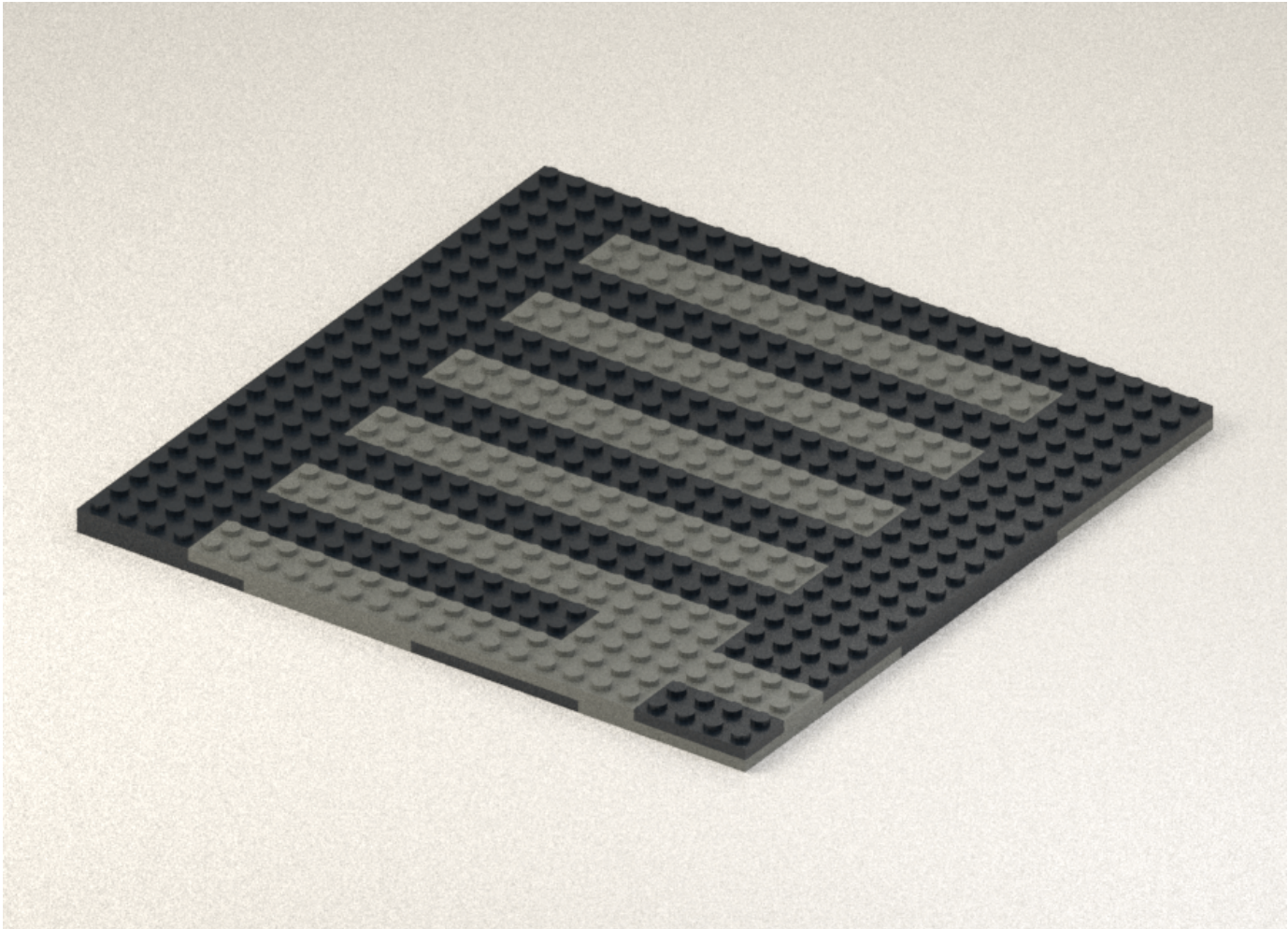
01 / Placement schedule

Place the pieces listed below; frame inserts identify their matching parent. Within a stage, assemble lower levels before higher levels unless the stage explicitly describes a subassembly.

Instance	Element	X, Y, Z	Footprint	Yaw / insert
P0001	303626	0, 0, 0	6 x 8	90 deg
P0002	4210794	6, 0, 0	6 x 8	90 deg
P0003	303626	12, 0, 0	6 x 8	90 deg
P0004	4210794	18, 0, 0	6 x 8	90 deg
P0005	4210794	0, 8, 0	6 x 8	90 deg
P0006	303626	6, 8, 0	6 x 8	90 deg
P0007	4210794	12, 8, 0	6 x 8	90 deg
P0008	303626	18, 8, 0	6 x 8	90 deg
P0009	303626	0, 16, 0	6 x 8	90 deg
P0010	4210794	6, 16, 0	6 x 8	90 deg
P0011	303626	12, 16, 0	6 x 8	90 deg
P0012	4210794	18, 16, 0	6 x 8	90 deg

02 / Lock the foundation

Add the offset upper layer. Complete it before lifting the base.



Add 42 pieces in this stage

Element	Color / geometry	Qty
302026	Black / 3020.dat	12
303426	Black / 3034.dat	17
4210997	Dark Bluish Gray / 3034.dat	13

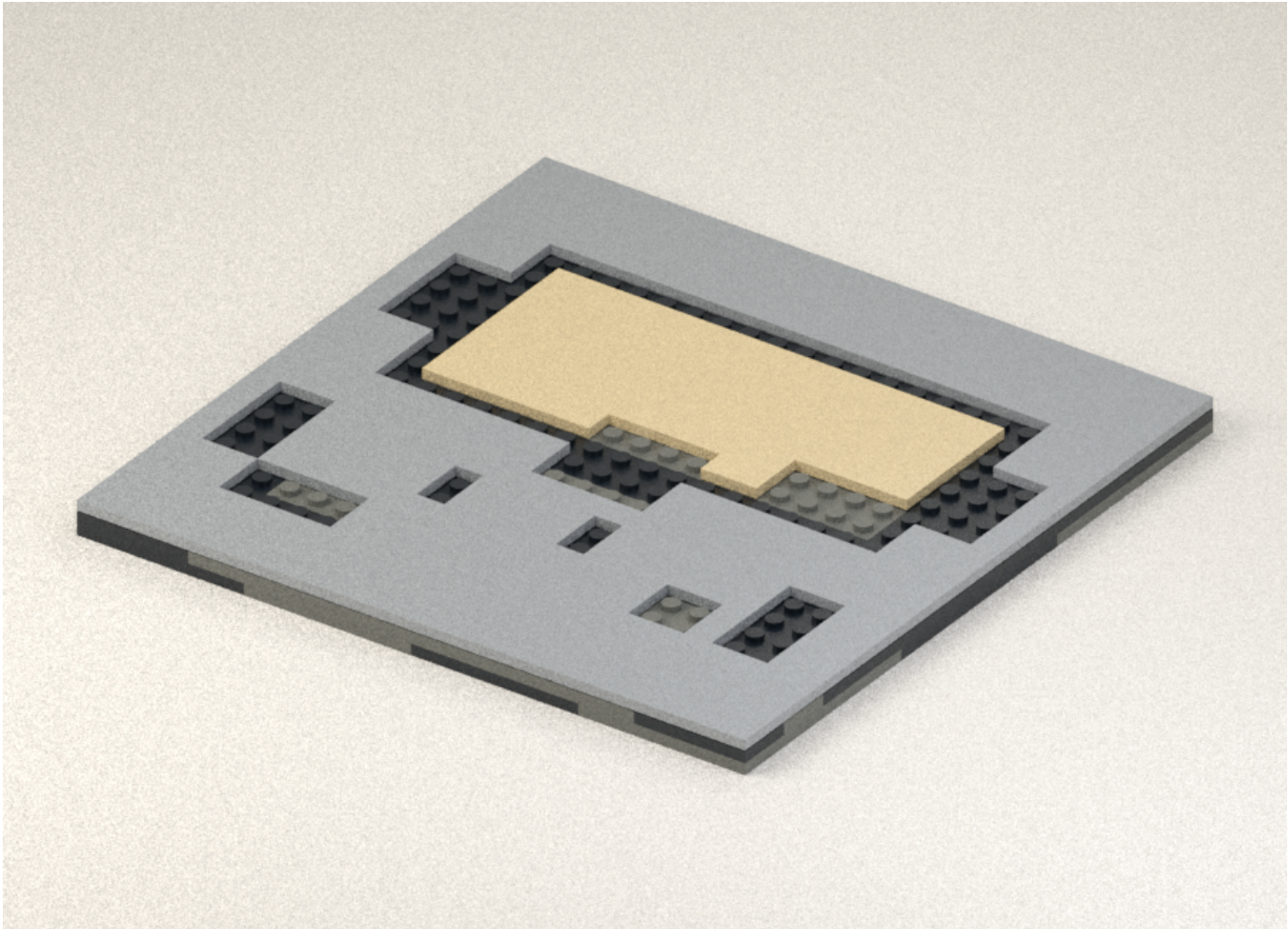
02 / Placement schedule

Place the pieces listed below; frame inserts identify their matching parent. Within a stage, assemble lower levels before higher levels unless the stage explicitly describes a subassembly.

Instance	Element	X, Y, Z	Footprint	Yaw / insert
P0013	303426	0, 0, 1	8 x 2	0 deg
P0014	303426	8, 0, 1	8 x 2	0 deg
P0015	303426	16, 0, 1	8 x 2	0 deg
P0016	4210997	4, 2, 1	8 x 2	0 deg
P0017	4210997	12, 2, 1	8 x 2	0 deg
P0018	302026	0, 2, 1	4 x 2	0 deg
P0019	302026	20, 2, 1	4 x 2	0 deg
P0020	303426	0, 4, 1	8 x 2	0 deg
P0021	303426	8, 4, 1	8 x 2	0 deg
P0022	303426	16, 4, 1	8 x 2	0 deg
P0023	4210997	4, 6, 1	8 x 2	0 deg
P0024	4210997	12, 6, 1	8 x 2	0 deg
P0025	302026	0, 6, 1	4 x 2	0 deg
P0026	302026	20, 6, 1	4 x 2	0 deg
P0027	303426	0, 8, 1	8 x 2	0 deg
P0028	303426	8, 8, 1	8 x 2	0 deg
P0029	303426	16, 8, 1	8 x 2	0 deg
P0030	4210997	4, 10, 1	8 x 2	0 deg
P0031	4210997	12, 10, 1	8 x 2	0 deg
P0032	302026	0, 10, 1	4 x 2	0 deg
P0033	302026	20, 10, 1	4 x 2	0 deg
P0034	303426	0, 12, 1	8 x 2	0 deg
P0035	303426	8, 12, 1	8 x 2	0 deg
P0036	303426	16, 12, 1	8 x 2	0 deg
P0037	4210997	4, 14, 1	8 x 2	0 deg
P0038	4210997	12, 14, 1	8 x 2	0 deg
P0039	302026	0, 14, 1	4 x 2	0 deg
P0040	302026	20, 14, 1	4 x 2	0 deg
P0041	303426	0, 16, 1	8 x 2	0 deg
P0042	303426	8, 16, 1	8 x 2	0 deg
P0043	303426	16, 16, 1	8 x 2	0 deg
P0044	4210997	4, 18, 1	8 x 2	0 deg
P0045	4210997	12, 18, 1	8 x 2	0 deg
P0046	302026	0, 18, 1	4 x 2	0 deg
P0047	302026	20, 18, 1	4 x 2	0 deg
P0048	303426	0, 20, 1	8 x 2	0 deg
P0049	303426	8, 20, 1	8 x 2	0 deg
P0050	4210997	16, 20, 1	8 x 2	0 deg
P0051	4210997	4, 22, 1	8 x 2	0 deg
P0052	4210997	12, 22, 1	8 x 2	0 deg
P0053	302026	0, 22, 1	4 x 2	0 deg
P0054	302026	20, 22, 1	4 x 2	0 deg

03 / Paved courtyard

Fit the pale-grey exterior and tan interior tiles. Leave the marked studded attachment areas.



Add 76 pieces in this stage

Element	Color / geometry	Qty
4114026	Tan / 3069b.dat	1
4157277	Tan / 6636.dat	1
4211356	Light Bluish Gray / 2431.dat	11
4211413	Light Bluish Gray / 3068b.dat	4
4211414	Light Bluish Gray / 3069b.dat	7
4211415	Light Bluish Gray / 3070b.dat	4
4211481	Light Bluish Gray / 4162.dat	13
4211549	Light Bluish Gray / 6636.dat	4
4550324	Tan / 2431.dat	1
4560183	Light Bluish Gray / 87079.dat	19
6122047	Tan / 87079.dat	11

03 / Placement schedule

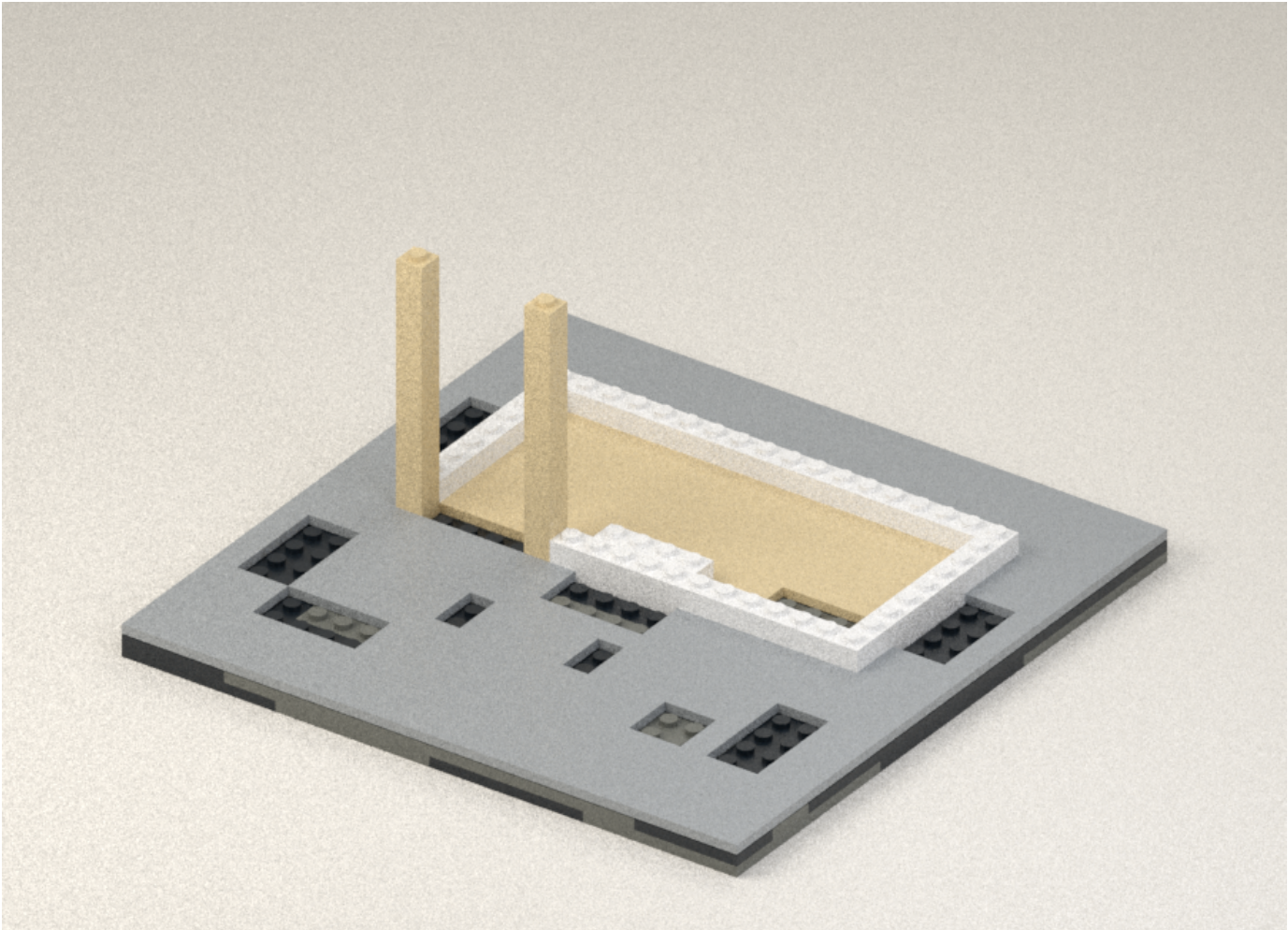
Place the pieces listed below; frame inserts identify their matching parent. Within a stage, assemble lower levels before higher levels unless the stage explicitly describes a subassembly.

Instance	Element	X, Y, Z	Footprint	Yaw / insert
P0055	4560183	0, 0, 2	4 x 2	0 deg
P0056	4560183	4, 0, 2	4 x 2	0 deg
P0057	4560183	8, 0, 2	4 x 2	0 deg
P0058	4560183	12, 0, 2	4 x 2	0 deg
P0059	4560183	16, 0, 2	4 x 2	0 deg
P0060	4560183	20, 0, 2	4 x 2	0 deg
P0061	4560183	0, 2, 2	4 x 2	0 deg
P0062	4560183	4, 2, 2	4 x 2	0 deg
P0063	4560183	8, 2, 2	4 x 2	0 deg
P0064	4560183	12, 2, 2	4 x 2	0 deg
P0065	4560183	16, 2, 2	4 x 2	0 deg
P0066	4560183	20, 2, 2	4 x 2	0 deg
P0067	4211481	0, 4, 2	1 x 8	90 deg
P0068	4211413	1, 4, 2	2 x 2	90 deg
P0069	4211413	21, 4, 2	2 x 2	90 deg
P0070	4211481	23, 4, 2	1 x 8	90 deg
P0071	4211414	21, 6, 2	2 x 1	0 deg
P0072	4560183	1, 10, 2	2 x 4	90 deg
P0073	4560183	21, 11, 2	2 x 4	90 deg
P0074	4211481	0, 12, 2	1 x 8	90 deg
P0075	4211481	23, 12, 2	1 x 8	90 deg
P0076	4560183	3, 13, 2	4 x 2	0 deg
P0077	4211481	7, 13, 2	1 x 8	90 deg
P0078	4211413	8, 13, 2	2 x 2	90 deg
P0079	4560183	14, 13, 2	4 x 2	0 deg
P0080	4560183	18, 13, 2	2 x 4	90 deg
P0081	4211414	20, 13, 2	1 x 2	90 deg
P0082	4211414	1, 14, 2	2 x 1	0 deg
P0083	4560183	3, 15, 2	4 x 2	0 deg
P0084	4211481	8, 15, 2	8 x 1	0 deg
P0085	4211481	16, 15, 2	1 x 8	90 deg
P0086	4211414	17, 15, 2	1 x 2	90 deg
P0087	4211414	20, 15, 2	2 x 1	0 deg
P0088	4211481	22, 15, 2	1 x 8	90 deg
P0089	4560183	9, 16, 2	4 x 2	0 deg
P0090	4211481	14, 16, 2	1 x 8	90 deg
P0091	4211481	15, 16, 2	1 x 8	90 deg
P0092	4211356	3, 17, 2	4 x 1	0 deg
P0093	4211414	17, 17, 2	2 x 1	0 deg
P0094	4211549	19, 17, 2	1 x 6	90 deg
P0095	4211356	3, 18, 2	4 x 1	0 deg
P0096	4211549	8, 18, 2	6 x 1	0 deg
P0097	4211356	1, 19, 2	1 x 4	90 deg
P0098	4211356	2, 19, 2	1 x 4	90 deg
P0099	4211549	8, 19, 2	6 x 1	0 deg
P0100	4211356	0, 20, 2	1 x 4	90 deg
P0101	4211549	8, 20, 2	6 x 1	0 deg
P0102	4211356	17, 20, 2	1 x 4	90 deg

Instance	Element	X, Y, Z	Footprint	Yaw / insert
P0103	4211356	18, 20, 2	1 x 4	90 deg
P0104	4211356	20, 20, 2	1 x 4	90 deg
P0105	4211356	21, 20, 2	1 x 4	90 deg
P0106	4211356	23, 20, 2	1 x 4	90 deg
P0107	4211481	3, 21, 2	8 x 1	0 deg
P0108	4211413	11, 21, 2	2 x 2	90 deg
P0109	4211414	13, 21, 2	1 x 2	90 deg
P0110	4211481	3, 22, 2	8 x 1	0 deg
P0111	4211481	1, 23, 2	8 x 1	0 deg
P0112	4211356	9, 23, 2	4 x 1	0 deg
P0113	4211415	13, 23, 2	1 x 1	90 deg
P0114	4211415	16, 23, 2	1 x 1	90 deg
P0115	4211415	19, 23, 2	1 x 1	90 deg
P0116	4211415	22, 23, 2	1 x 1	90 deg
P0117	6122047	4, 5, 2	4 x 2	0 deg
P0118	6122047	8, 5, 2	4 x 2	0 deg
P0119	6122047	12, 5, 2	4 x 2	0 deg
P0120	6122047	16, 5, 2	4 x 2	0 deg
P0121	6122047	4, 7, 2	4 x 2	0 deg
P0122	6122047	8, 7, 2	4 x 2	0 deg
P0123	6122047	12, 7, 2	4 x 2	0 deg
P0124	6122047	16, 7, 2	4 x 2	0 deg
P0125	6122047	4, 9, 2	4 x 2	0 deg
P0126	6122047	8, 9, 2	4 x 2	0 deg
P0127	6122047	12, 9, 2	4 x 2	0 deg
P0128	4550324	16, 9, 2	4 x 1	0 deg
P0129	4157277	4, 11, 2	6 x 1	0 deg
P0130	4114026	14, 11, 2	2 x 1	0 deg

04 / Window sills and pillars

Build the low sills and tan entrance pillars.



Add 26 pieces in this stage

Element	Color / geometry	Qty
300901	White / 3009.dat	4
301001	White / 3010.dat	6
4113915	Tan / 3005.dat	16

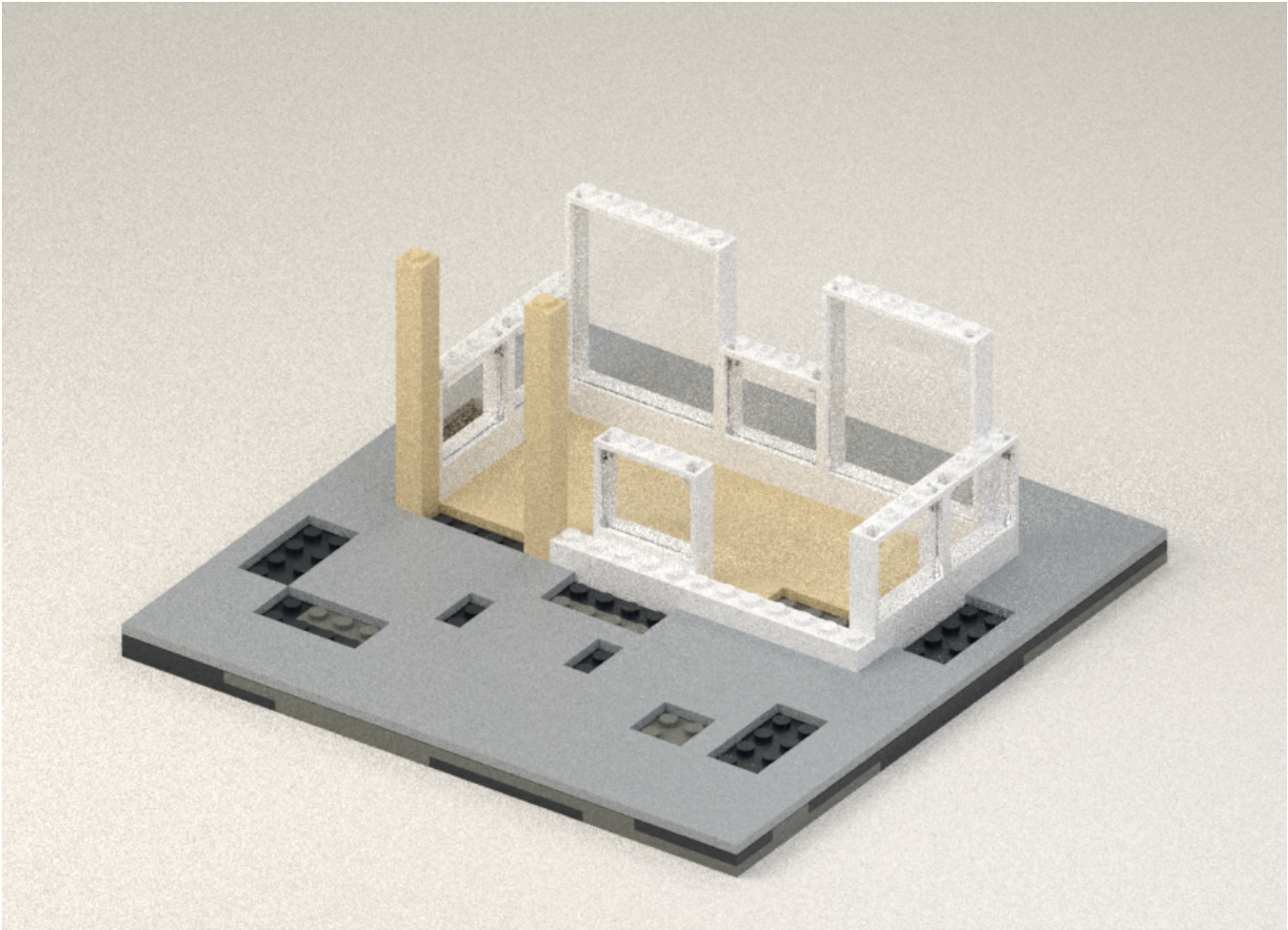
04 / Placement schedule

Place the pieces listed below; frame inserts identify their matching parent. Within a stage, assemble lower levels before higher levels unless the stage explicitly describes a subassembly.

Instance	Element	X, Y, Z	Footprint	Yaw / insert
P0131	4113915	3, 12, 2	1 x 1	0 deg
P0132	4113915	8, 12, 2	1 x 1	0 deg
P0133	4113915	3, 12, 5	1 x 1	0 deg
P0134	4113915	8, 12, 5	1 x 1	0 deg
P0135	4113915	3, 12, 8	1 x 1	0 deg
P0136	4113915	8, 12, 8	1 x 1	0 deg
P0137	4113915	3, 12, 11	1 x 1	0 deg
P0138	4113915	8, 12, 11	1 x 1	0 deg
P0139	4113915	3, 12, 14	1 x 1	0 deg
P0140	4113915	8, 12, 14	1 x 1	0 deg
P0141	4113915	3, 12, 17	1 x 1	0 deg
P0142	4113915	8, 12, 17	1 x 1	0 deg
P0143	4113915	3, 12, 20	1 x 1	0 deg
P0144	4113915	8, 12, 20	1 x 1	0 deg
P0145	4113915	3, 12, 23	1 x 1	0 deg
P0146	4113915	8, 12, 23	1 x 1	0 deg
P0147	300901	9, 12, 2	6 x 1	0 deg
P0148	300901	15, 12, 2	6 x 1	0 deg
P0149	301001	3, 4, 2	1 x 4	90 deg
P0150	301001	3, 8, 2	1 x 4	90 deg
P0151	301001	20, 4, 2	1 x 4	90 deg
P0152	301001	20, 8, 2	1 x 4	90 deg
P0153	300901	4, 4, 2	6 x 1	0 deg
P0154	300901	14, 4, 2	6 x 1	0 deg
P0155	301001	10, 4, 2	4 x 1	0 deg
P0156	301001	10, 11, 2	4 x 1	0 deg

05 / Lower glazing

Fit the lower side and rear frames with their matching panes.



Add 16 pieces in this stage

Element	Color / geometry	Qty
4530590	White / 60594.dat	6
6256127	White / 42205.dat	2
6511024	Trans-Clear / 42509.dat	2
6514142	Trans-Clear / 86210.dat	6

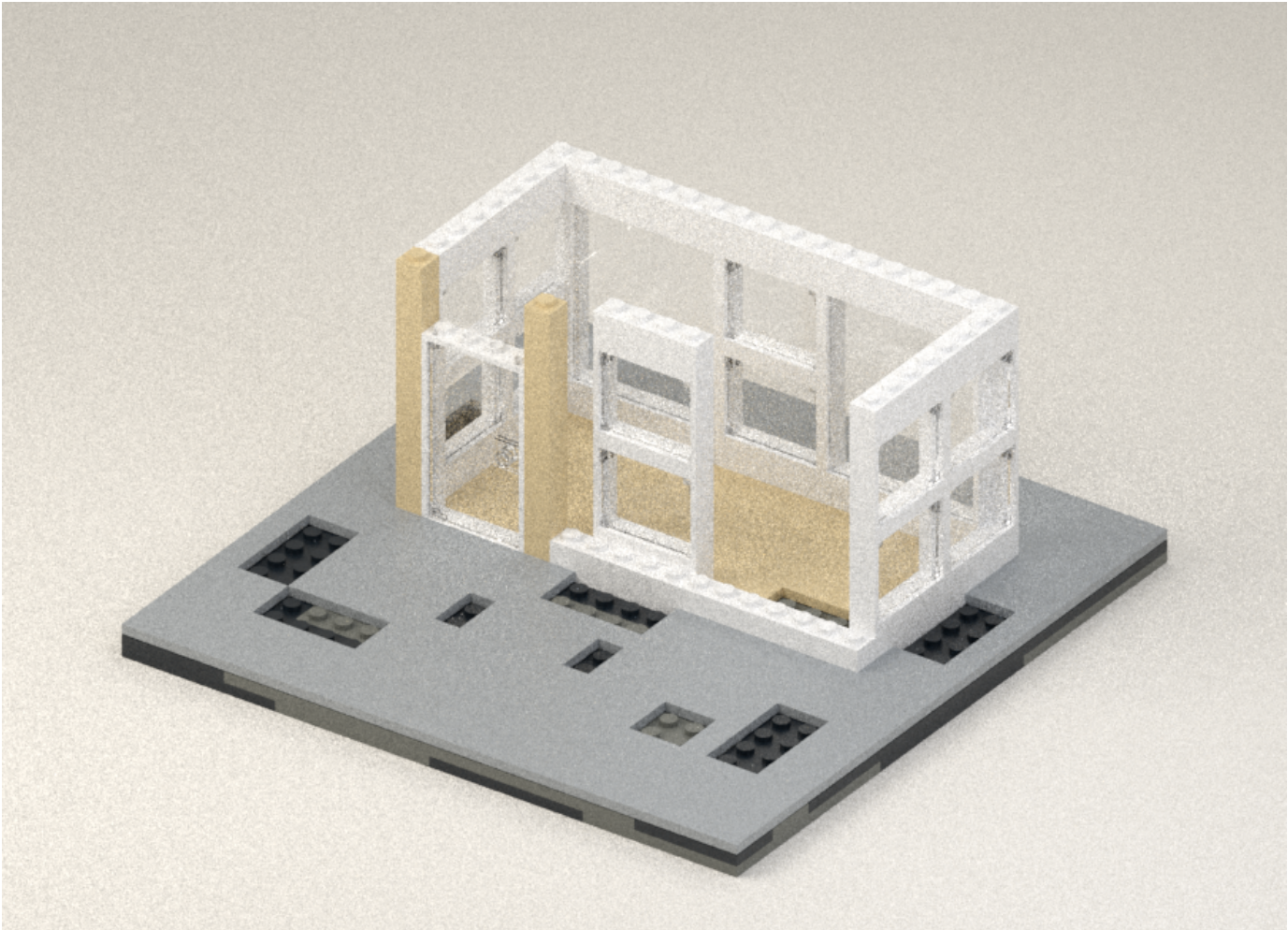
05 / Placement schedule

Place the pieces listed below; frame inserts identify their matching parent. Within a stage, assemble lower levels before higher levels unless the stage explicitly describes a subassembly.

Instance	Element	X, Y, Z	Footprint	Yaw / insert
P0157	4530590	3, 4, 5	1 x 4	90 deg
P0158	6514142	Frame P0157	Insert	offset 0, 8, 4
P0162	4530590	3, 8, 5	1 x 4	90 deg
P0163	6514142	Frame P0162	Insert	offset 0, 8, 4
P0167	4530590	20, 4, 5	1 x 4	90 deg
P0168	6514142	Frame P0167	Insert	offset 0, 8, 4
P0172	4530590	20, 8, 5	1 x 4	90 deg
P0173	6514142	Frame P0172	Insert	offset 0, 8, 4
P0177	6256127	4, 4, 5	6 x 1	0 deg
P0178	6511024	Frame P0177	Insert	offset 0, 4.5, 5
P0180	6256127	14, 4, 5	6 x 1	0 deg
P0181	6511024	Frame P0180	Insert	offset 0, 4.5, 5
P0183	4530590	10, 4, 5	4 x 1	0 deg
P0184	6514142	Frame P0183	Insert	offset 0, 8, 4
P0188	4530590	10, 11, 5	4 x 1	0 deg
P0189	6514142	Frame P0188	Insert	offset 0, 8, 4

06 / Upper glazing and entrance

Add the upper windows and the glazed entrance door. Insert panes before the cornice.



Add 22 pieces in this stage

Element	Color / geometry	Qty
300901	White / 3009.dat	2
301001	White / 3010.dat	6
4530590	White / 60594.dat	6
6262945	White / 60596.dat	1
6514119	Trans-Clear / 60616a.dat	1
6514142	Trans-Clear / 86210.dat	6

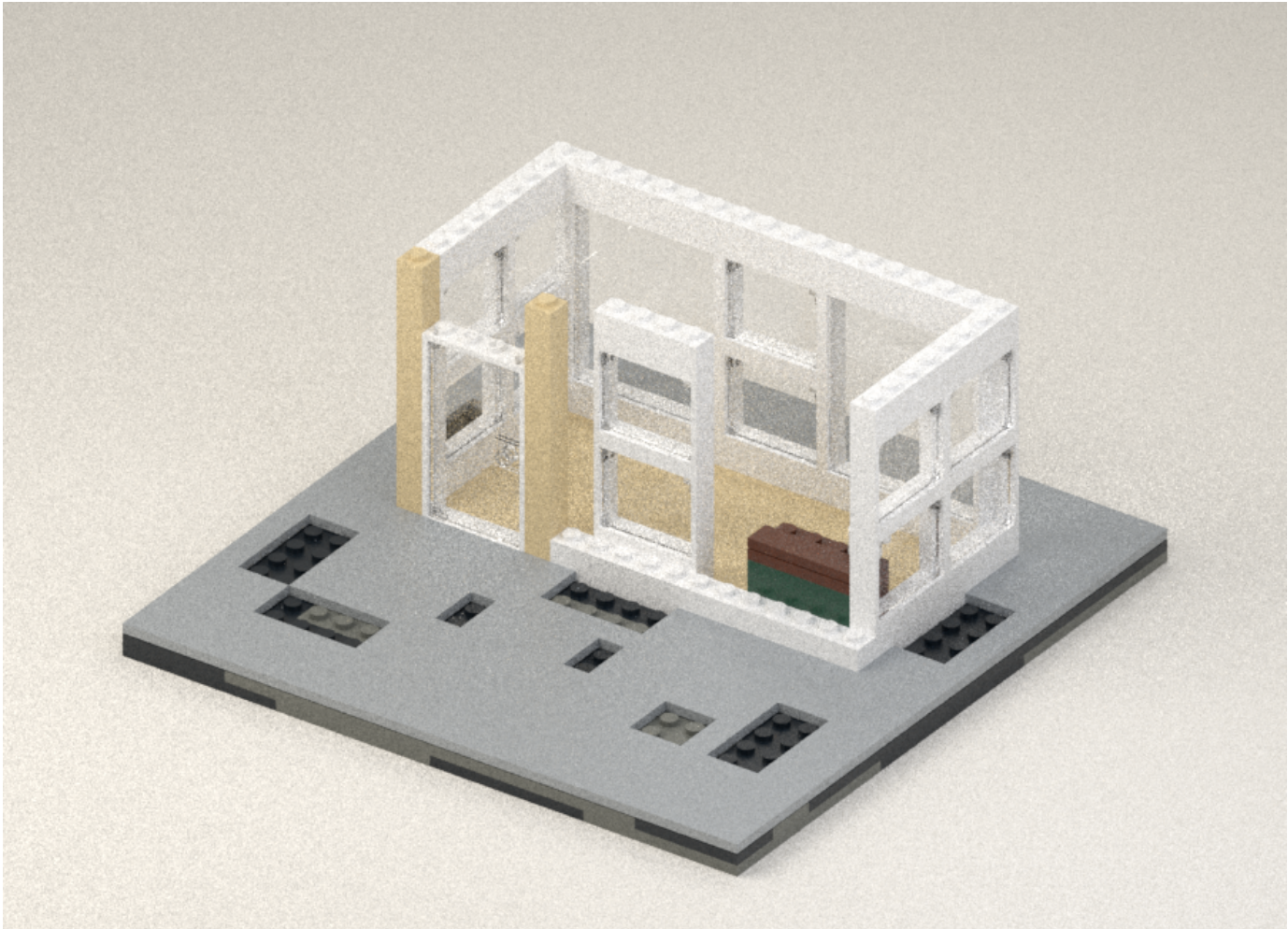
06 / Placement schedule

Place the pieces listed below; frame inserts identify their matching parent. Within a stage, assemble lower levels before higher levels unless the stage explicitly describes a subassembly.

Instance	Element	X, Y, Z	Footprint	Yaw / insert
P0159	4530590	3, 4, 14	1 x 4	90 deg
P0160	6514142	Frame P0159	Insert	offset 0, 8, 4
P0161	301001	3, 4, 23	1 x 4	90 deg
P0164	4530590	3, 8, 14	1 x 4	90 deg
P0165	6514142	Frame P0164	Insert	offset 0, 8, 4
P0166	301001	3, 8, 23	1 x 4	90 deg
P0169	4530590	20, 4, 14	1 x 4	90 deg
P0170	6514142	Frame P0169	Insert	offset 0, 8, 4
P0171	301001	20, 4, 23	1 x 4	90 deg
P0174	4530590	20, 8, 14	1 x 4	90 deg
P0175	6514142	Frame P0174	Insert	offset 0, 8, 4
P0176	301001	20, 8, 23	1 x 4	90 deg
P0179	300901	4, 4, 23	6 x 1	0 deg
P0182	300901	14, 4, 23	6 x 1	0 deg
P0185	4530590	10, 4, 14	4 x 1	0 deg
P0186	6514142	Frame P0185	Insert	offset 0, 8, 4
P0187	301001	10, 4, 23	4 x 1	0 deg
P0190	4530590	10, 11, 14	4 x 1	0 deg
P0191	6514142	Frame P0190	Insert	offset 0, 8, 4
P0192	301001	10, 11, 23	4 x 1	0 deg
P0193	6262945	4, 12, 2	4 x 1	0 deg
P0194	6514119	Frame P0193	Insert	offset -32, 0, 5

07 / Cafe counter

Build the counter inside the open serving arch, then add its white mug.



Add 8 pieces in this stage

Element	Color / geometry	Qty
4211151	Reddish Brown / 3069b.dat	2
4211186	Reddish Brown / 3020.dat	1
4211225	Reddish Brown / 3010.dat	2
4245571	Dark Green / 3010.dat	2
4659665	White / 3899.dat	1

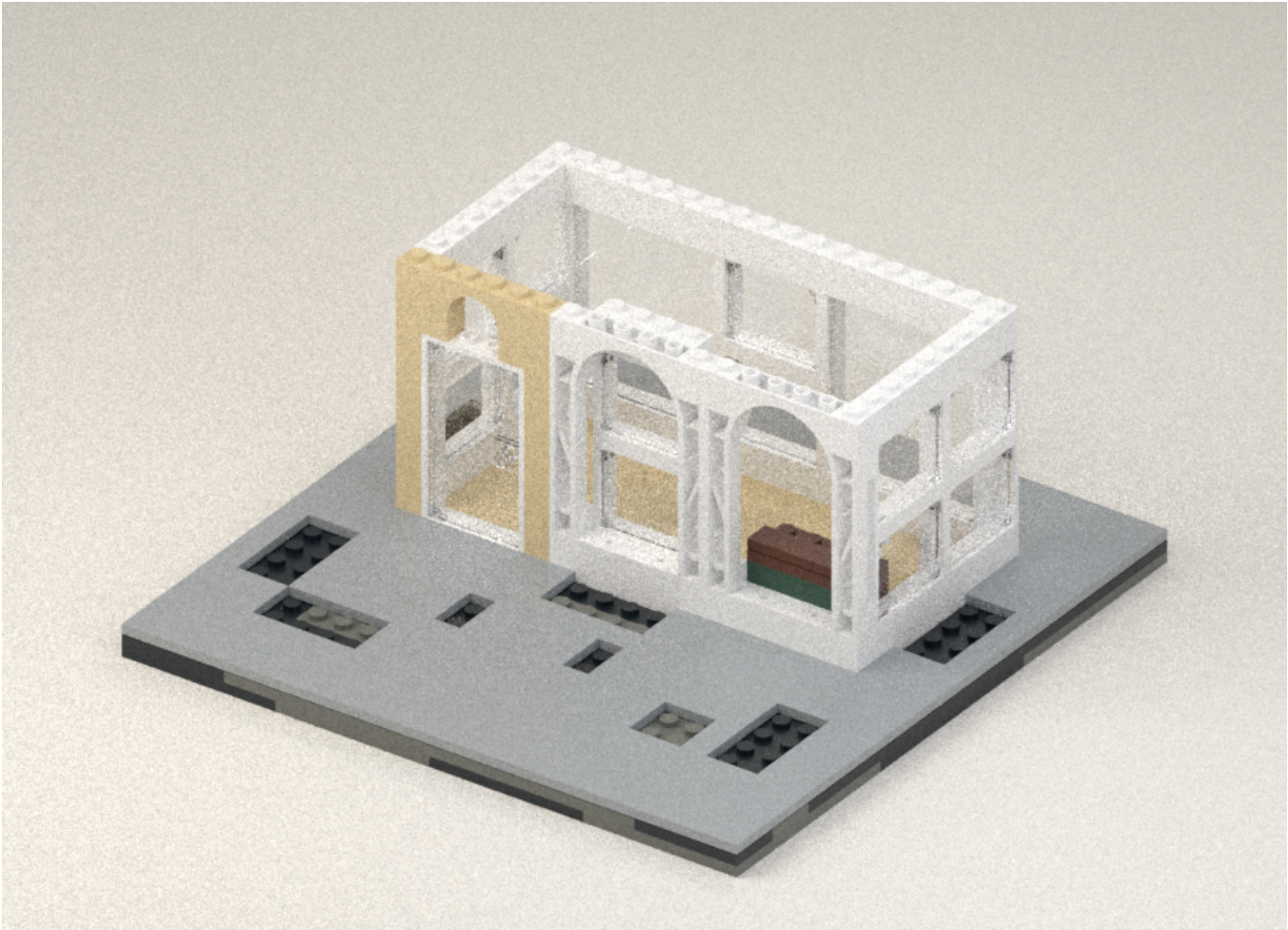
07 / Placement schedule

Place the pieces listed below; frame inserts identify their matching parent. Within a stage, assemble lower levels before higher levels unless the stage explicitly describes a subassembly.

Instance	Element	X, Y, Z	Footprint	Yaw / insert
P0277	4211225	16, 10, 2	4 x 1	0 deg
P0278	4245571	16, 11, 2	4 x 1	0 deg
P0279	4211225	16, 10, 5	4 x 1	0 deg
P0280	4245571	16, 11, 5	4 x 1	0 deg
P0281	4211186	16, 10, 8	4 x 2	0 deg
P0282	4211151	16, 11, 9	2 x 1	0 deg
P0283	4211151	18, 11, 9	2 x 1	0 deg
P0284	4659665	19, 10, 9	1 x 1	180 deg

08 / Arched facade

Add both large arches, the small tan entrance arch, and the four outer-pillar fillers.



Add 7 pieces in this stage

Element	Color / geometry	Qty
300501	White / 3005.dat	4
6031057	Tan / 6182.dat	1
6249033	White / 40066.dat	2

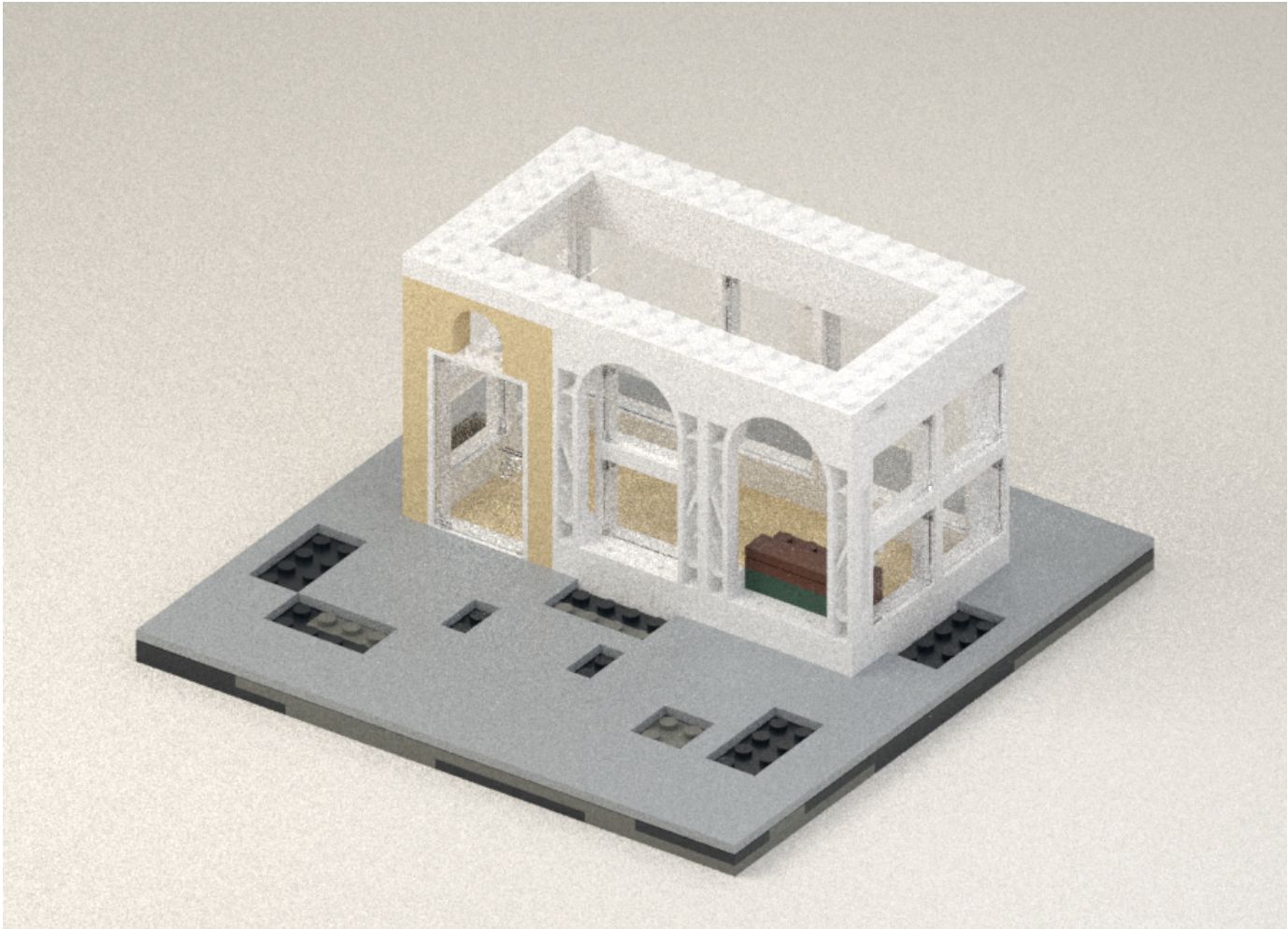
08 / Placement schedule

Place the pieces listed below; frame inserts identify their matching parent. Within a stage, assemble lower levels before higher levels unless the stage explicitly describes a subassembly.

Instance	Element	X, Y, Z	Footprint	Yaw / insert
P0195	6031057	4, 12, 20	4 x 1	0 deg
P0196	6249033	9, 12, 5	6 x 1	0 deg
P0197	300501	9, 12, 23	1 x 1	0 deg
P0198	300501	14, 12, 23	1 x 1	0 deg
P0199	6249033	15, 12, 5	6 x 1	0 deg
P0200	300501	15, 12, 23	1 x 1	0 deg
P0201	300501	20, 12, 23	1 x 1	0 deg

09 / Roof cornice

Tie the walls with two crossing plate layers. Keep the central glazed opening clear.



Add 16 pieces in this stage

Element	Color / geometry	Qty
366601	White / 3666.dat	8
379501	White / 3795.dat	8

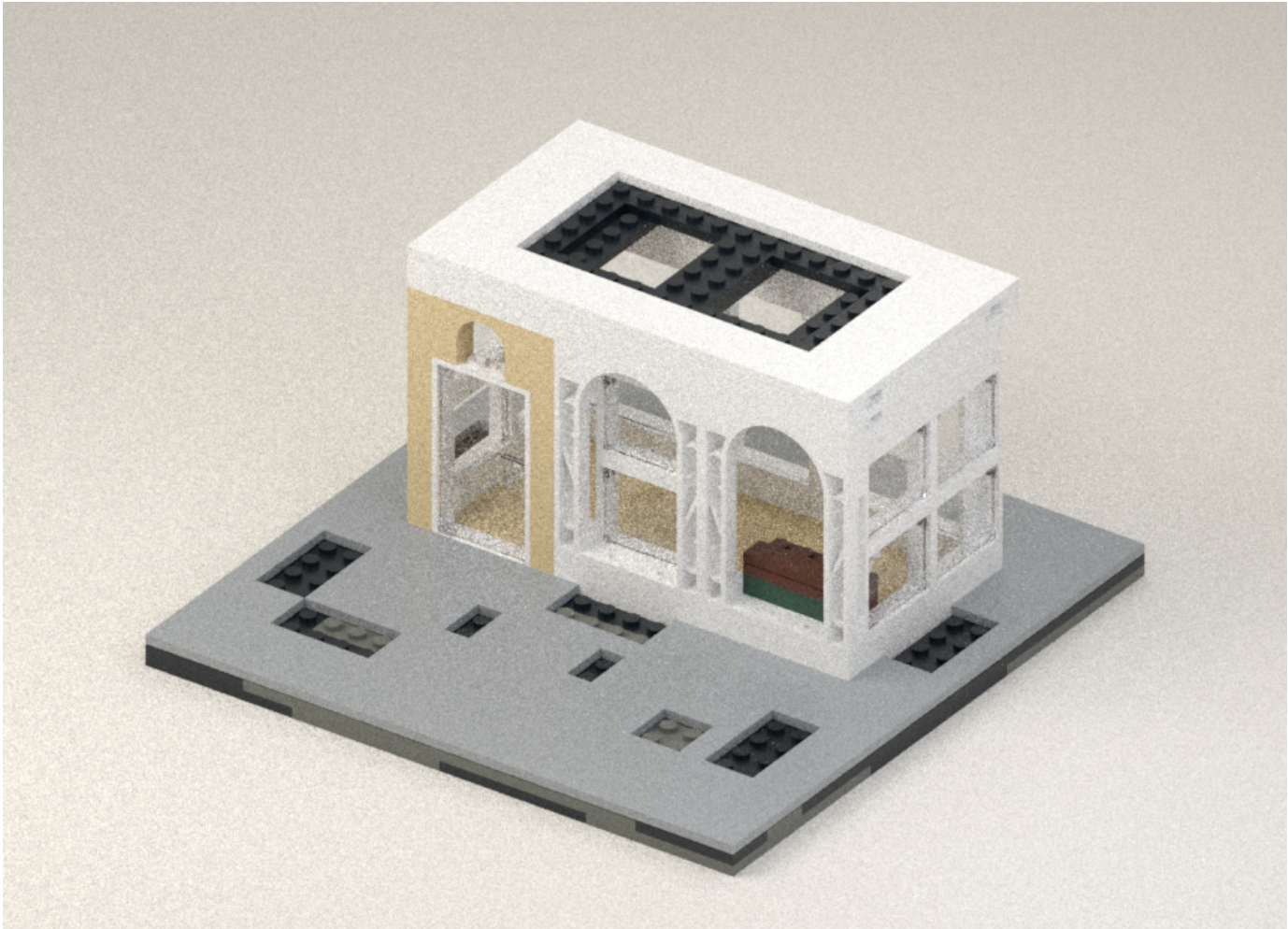
09 / Placement schedule

Place the pieces listed below; frame inserts identify their matching parent. Within a stage, assemble lower levels before higher levels unless the stage explicitly describes a subassembly.

Instance	Element	X, Y, Z	Footprint	Yaw / insert
P0202	366601	3, 4, 26	6 x 1	0 deg
P0203	366601	9, 4, 26	6 x 1	0 deg
P0204	366601	15, 4, 26	6 x 1	0 deg
P0205	366601	3, 12, 26	6 x 1	0 deg
P0206	366601	9, 12, 26	6 x 1	0 deg
P0207	366601	15, 12, 26	6 x 1	0 deg
P0208	366601	3, 5, 26	1 x 6	90 deg
P0209	366601	20, 5, 26	1 x 6	90 deg
P0210	379501	3, 3, 27	6 x 2	0 deg
P0211	379501	9, 3, 27	6 x 2	0 deg
P0212	379501	15, 3, 27	6 x 2	0 deg
P0213	379501	3, 11, 27	6 x 2	0 deg
P0214	379501	9, 11, 27	6 x 2	0 deg
P0215	379501	15, 11, 27	6 x 2	0 deg
P0216	379501	3, 5, 27	2 x 6	90 deg
P0217	379501	19, 5, 27	2 x 6	90 deg

10 / Lantern support frame

Make the lantern rail subassembly: connect the three lower beams with the two eave rows before placing it on the cornice. Add the upper rails and shoulder trim.



Add 43 pieces in this stage

Element	Color / geometry	Qty
302326	Black / 3023b.dat	4
366601	White / 3666.dat	8
371026	Black / 3710.dat	4
379501	White / 3795.dat	2
379526	Black / 3795.dat	7
663601	White / 6636.dat	18

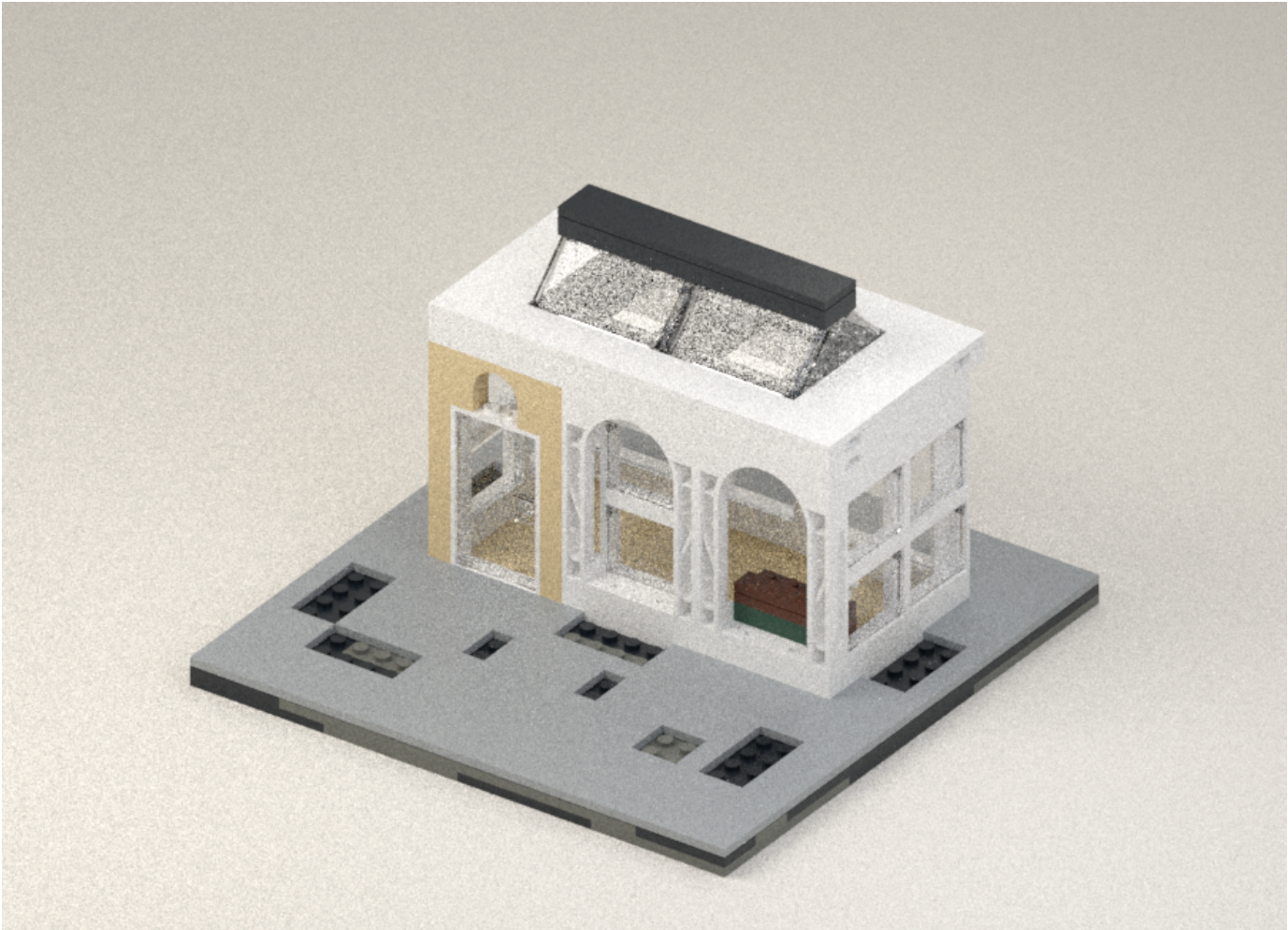
10 / Placement schedule

Place the pieces listed below; frame inserts identify their matching parent. Within a stage, assemble lower levels before higher levels unless the stage explicitly describes a subassembly.

Instance	Element	X, Y, Z	Footprint	Yaw / insert
P0218	366601	3, 5, 28	1 x 6	90 deg
P0219	379501	4, 5, 28	2 x 6	90 deg
P0220	379501	18, 5, 28	2 x 6	90 deg
P0221	366601	20, 5, 28	1 x 6	90 deg
P0222	366601	3, 3, 28	6 x 1	0 deg
P0223	366601	9, 3, 28	6 x 1	0 deg
P0224	366601	15, 3, 28	6 x 1	0 deg
P0225	366601	3, 12, 28	6 x 1	0 deg
P0226	366601	9, 12, 28	6 x 1	0 deg
P0227	366601	15, 12, 28	6 x 1	0 deg
P0228	302326	3, 4, 28	2 x 1	0 deg
P0229	302326	3, 11, 28	2 x 1	0 deg
P0230	302326	18, 4, 28	2 x 1	0 deg
P0231	302326	18, 11, 28	2 x 1	0 deg
P0232	379526	6, 4, 28	6 x 2	0 deg
P0233	379526	12, 4, 28	6 x 2	0 deg
P0234	379526	6, 10, 28	6 x 2	0 deg
P0235	379526	12, 10, 28	6 x 2	0 deg
P0236	371026	6, 6, 28	1 x 4	90 deg
P0237	371026	17, 6, 28	1 x 4	90 deg
P0238	379526	6, 5, 27	2 x 6	90 deg
P0239	379526	11, 5, 27	2 x 6	90 deg
P0240	379526	16, 5, 27	2 x 6	90 deg
P0241	371026	11, 6, 28	1 x 4	90 deg
P0242	371026	12, 6, 28	1 x 4	90 deg
P0243	663601	3, 3, 29	6 x 1	0 deg
P0244	663601	9, 3, 29	6 x 1	0 deg
P0245	663601	15, 3, 29	6 x 1	0 deg
P0246	663601	3, 4, 29	6 x 1	0 deg
P0247	663601	9, 4, 29	6 x 1	0 deg
P0248	663601	15, 4, 29	6 x 1	0 deg
P0249	663601	3, 11, 29	6 x 1	0 deg
P0250	663601	9, 11, 29	6 x 1	0 deg
P0251	663601	15, 11, 29	6 x 1	0 deg
P0252	663601	3, 12, 29	6 x 1	0 deg
P0253	663601	9, 12, 29	6 x 1	0 deg
P0254	663601	15, 12, 29	6 x 1	0 deg
P0255	663601	3, 5, 29	1 x 6	90 deg
P0256	663601	4, 5, 29	1 x 6	90 deg
P0257	663601	5, 5, 29	1 x 6	90 deg
P0258	663601	18, 5, 29	1 x 6	90 deg
P0259	663601	19, 5, 29	1 x 6	90 deg
P0260	663601	20, 5, 29	1 x 6	90 deg

11 / Glass roof

Fit the four fixed windscreens in opposing pairs; cap and tie their upper studs.



Add 16 pieces in this stage

Element	Color / geometry	Qty
371026	Black / 3710.dat	6
379526	Black / 3795.dat	2
6503221	Trans-Clear / 92583.dat	4
663626	Black / 6636.dat	4

11 / Placement schedule

Place the pieces listed below; frame inserts identify their matching parent. Within a stage, assemble lower levels before higher levels unless the stage explicitly describes a subassembly.

Instance	Element	X, Y, Z	Footprint	Yaw / insert
P0261	6503221	6, 5, 29	6 x 3	0 deg
P0262	6503221	6, 8, 29	6 x 3	180 deg
P0263	6503221	12, 5, 29	6 x 3	0 deg
P0264	6503221	12, 8, 29	6 x 3	180 deg
P0265	379526	6, 7, 35	6 x 2	0 deg
P0266	379526	12, 7, 35	6 x 2	0 deg
P0267	371026	6, 7, 36	4 x 1	0 deg
P0268	371026	10, 7, 36	4 x 1	0 deg
P0269	371026	14, 7, 36	4 x 1	0 deg
P0270	663626	6, 7, 37	6 x 1	0 deg
P0271	663626	12, 7, 37	6 x 1	0 deg
P0272	371026	6, 8, 36	4 x 1	0 deg
P0273	371026	10, 8, 36	4 x 1	0 deg
P0274	371026	14, 8, 36	4 x 1	0 deg
P0275	663626	6, 8, 37	6 x 1	0 deg
P0276	663626	12, 8, 37	6 x 1	0 deg

12 / Bench and bistro table

Build the timber bench and compact square table.



Add 10 pieces in this stage

Element	Color / geometry	Qty
300301	White / 3003.dat	1
4211149	Reddish Brown / 3004.dat	3
4211204	Reddish Brown / 6636.dat	2
4211247	Reddish Brown / 3795.dat	1
4211388	Light Bluish Gray / 3004.dat	2
4216695	Reddish Brown / 3022.dat	1

12 / Placement schedule

Place the pieces listed below; frame inserts identify their matching parent. Within a stage, assemble lower levels before higher levels unless the stage explicitly describes a subassembly.

Instance	Element	X, Y, Z	Footprint	Yaw / insert
P0285	4211388	8, 16, 2	1 x 2	90 deg
P0286	4211388	13, 16, 2	1 x 2	90 deg
P0287	4211247	8, 16, 5	6 x 2	0 deg
P0288	4211149	8, 16, 6	2 x 1	0 deg
P0289	4211149	12, 16, 6	2 x 1	0 deg
P0290	4211149	10, 16, 6	2 x 1	0 deg
P0291	4211204	8, 16, 9	6 x 1	0 deg
P0292	4211204	8, 17, 6	6 x 1	0 deg
P0293	300301	17, 18, 2	2 x 2	0 deg
P0294	4216695	17, 18, 5	2 x 2	0 deg

13 / Raised garden beds

Add the brown raised beds and brown mounting plates.



Add 13 pieces in this stage

Element	Color / geometry	Qty
4211186	Reddish Brown / 3020.dat	6
4211201	Reddish Brown / 3001.dat	5
4211225	Reddish Brown / 3010.dat	2

13 / Placement schedule

Place the pieces listed below; frame inserts identify their matching parent. Within a stage, assemble lower levels before higher levels unless the stage explicitly describes a subassembly.

Instance	Element	X, Y, Z	Footprint	Yaw / insert
P0296	4211201	1, 15, 2	2 x 4	90 deg
P0297	4211186	1, 15, 5	2 x 4	90 deg
P0306	4211201	3, 19, 2	4 x 2	0 deg
P0307	4211186	3, 19, 5	4 x 2	0 deg
P0314	4211201	20, 16, 2	2 x 4	90 deg
P0315	4211186	20, 16, 5	2 x 4	90 deg
P0324	4211225	10, 13, 2	4 x 1	0 deg
P0325	4211225	10, 14, 2	4 x 1	0 deg
P0326	4211186	10, 13, 5	4 x 2	0 deg
P0333	4211201	1, 6, 2	2 x 4	90 deg
P0334	4211186	1, 6, 5	2 x 4	90 deg
P0343	4211201	21, 7, 2	2 x 4	90 deg
P0344	4211186	21, 7, 5	2 x 4	90 deg

14 / Flowers and finishing

Attach leaves and flowers at the recorded studs; finish with the table mug.



Add 45 pieces in this stage

Element	Color / geometry	Qty
4211183	Reddish Brown / 3062b.dat	8
4659665	White / 3899.dat	1
6182261	Bright Green / 32607.dat	4
6206149	White / 24866.dat	4
6212501	Bright Pink / 24866.dat	4
6229130	Green / 32607.dat	12
6295242	Coral / 24866.dat	4
6400754	Dark Green / 32607.dat	4
6447541	Dark Green / 2682a.dat	4

14 / Placement schedule

Place the pieces listed below; frame inserts identify their matching parent. Within a stage, assemble lower levels before higher levels unless the stage explicitly describes a subassembly.

Instance	Element	X, Y, Z	Footprint	Yaw / insert
P0295	4659665	18, 19, 6	1 x 1	0 deg
P0298	4211183	1, 15, 6	1 x 1	0 deg
P0299	6400754	1, 15, 9	1 x 1	0 deg
P0300	6212501	1, 15, 10	1 x 1	0 deg
P0301	6447541	2, 16, 6	1 x 1	90 deg
P0302	4211183	1, 17, 6	1 x 1	0 deg
P0303	6229130	1, 17, 9	1 x 1	0 deg
P0304	6206149	1, 17, 10	1 x 1	0 deg
P0305	6229130	2, 18, 6	1 x 1	180 deg
P0308	6229130	3, 19, 6	1 x 1	0 deg
P0309	6206149	3, 19, 7	1 x 1	0 deg
P0310	6229130	4, 20, 6	1 x 1	180 deg
P0311	6182261	5, 19, 6	1 x 1	0 deg
P0312	6295242	5, 19, 7	1 x 1	0 deg
P0313	6229130	6, 20, 6	1 x 1	180 deg
P0316	4211183	20, 16, 6	1 x 1	0 deg
P0317	6182261	20, 16, 9	1 x 1	0 deg
P0318	6295242	20, 16, 10	1 x 1	0 deg
P0319	6447541	21, 17, 6	1 x 1	90 deg
P0320	4211183	20, 18, 6	1 x 1	0 deg
P0321	6400754	20, 18, 9	1 x 1	0 deg
P0322	6212501	20, 18, 10	1 x 1	0 deg
P0323	6229130	21, 19, 6	1 x 1	180 deg
P0327	6400754	10, 13, 6	1 x 1	180 deg
P0328	6212501	10, 13, 7	1 x 1	180 deg
P0329	6229130	11, 14, 6	1 x 1	270 deg
P0330	6229130	12, 13, 6	1 x 1	0 deg
P0331	6206149	12, 13, 7	1 x 1	0 deg
P0332	6229130	13, 14, 6	1 x 1	270 deg
P0335	4211183	1, 6, 6	1 x 1	0 deg
P0336	6229130	1, 6, 9	1 x 1	180 deg
P0337	6206149	1, 6, 10	1 x 1	180 deg
P0338	6447541	2, 6, 6	1 x 1	180 deg
P0339	4211183	1, 8, 6	1 x 1	0 deg
P0340	6182261	1, 8, 9	1 x 1	180 deg
P0341	6295242	1, 8, 10	1 x 1	180 deg
P0342	6229130	2, 9, 6	1 x 1	180 deg
P0345	4211183	21, 7, 6	1 x 1	0 deg
P0346	6182261	21, 7, 9	1 x 1	0 deg
P0347	6295242	21, 7, 10	1 x 1	0 deg
P0348	6447541	22, 9, 6	1 x 1	0 deg
P0349	4211183	21, 9, 6	1 x 1	0 deg
P0350	6400754	21, 9, 9	1 x 1	0 deg
P0351	6212501	21, 9, 10	1 x 1	0 deg
P0352	6229130	22, 7, 6	1 x 1	0 deg

Combined bill of materials

Exact element IDs distinguish both shape and color. Available means the conservative set-membership allowance. See [picking-list.md](#) and [allocation.json](#) for donor boxes.

Element	Color / geometry	Need	Available
300301	White / 3003.dat	1	19
300501	White / 3005.dat	4	43
300901	White / 3009.dat	6	22
301001	White / 3010.dat	12	38
302026	Black / 3020.dat	12	34
302326	Black / 3023b.dat	4	43
303426	Black / 3034.dat	17	17
303626	Black / 3036.dat	6	6
366601	White / 3666.dat	16	36
371026	Black / 3710.dat	10	29
379501	White / 3795.dat	10	23
379526	Black / 3795.dat	9	25
4113915	Tan / 3005.dat	16	17
4114026	Tan / 3069b.dat	1	20
4157277	Tan / 6636.dat	1	9
4210794	Dark Bluish Gray / 3036.dat	6	7
4210997	Dark Bluish Gray / 3034.dat	13	13
4211149	Reddish Brown / 3004.dat	3	10
4211151	Reddish Brown / 3069b.dat	2	7
4211183	Reddish Brown / 3062b.dat	8	9
4211186	Reddish Brown / 3020.dat	7	17
4211201	Reddish Brown / 3001.dat	5	5
4211204	Reddish Brown / 6636.dat	2	8
4211225	Reddish Brown / 3010.dat	4	7
4211247	Reddish Brown / 3795.dat	1	6
4211356	Light Bluish Gray / 2431.dat	11	23
4211388	Light Bluish Gray / 3004.dat	2	19
4211413	Light Bluish Gray / 3068b.dat	4	19
4211414	Light Bluish Gray / 3069b.dat	7	23
4211415	Light Bluish Gray / 3070b.dat	4	14
4211481	Light Bluish Gray / 4162.dat	13	15
4211549	Light Bluish Gray / 6636.dat	4	13
4216695	Reddish Brown / 3022.dat	1	13
4245571	Dark Green / 3010.dat	2	3
4530590	White / 60594.dat	12	13
4550324	Tan / 2431.dat	1	14
4560183	Light Bluish Gray / 87079.dat	19	19
4659665	White / 3899.dat	2	4
6031057	Tan / 6182.dat	1	2
6122047	Tan / 87079.dat	11	11
6182261	Bright Green / 32607.dat	4	28
6206149	White / 24866.dat	4	8
6212501	Bright Pink / 24866.dat	4	13
6229130	Green / 32607.dat	12	20
6249033	White / 40066.dat	2	2
6256127	White / 42205.dat	2	12

Element	Color / geometry	Need	Available
6262945	White / 60596.dat	1	15
6295242	Coral / 24866.dat	4	13
6400754	Dark Green / 32607.dat	4	11
6447541	Dark Green / 2682a.dat	4	4
6503221	Trans-Clear / 92583.dat	4	5
6511024	Trans-Clear / 42509.dat	2	5
6514119	Trans-Clear / 60616a.dat	1	12
6514142	Trans-Clear / 86210.dat	12	13
663601	White / 6636.dat	18	28
663626	Black / 6636.dat	4	28

Provenance

Inventory: repo data/owned-sets.csv and public/data/parts/*.json, frozen with input hashes. Exact part/color mapping: installed BrickLink Studio elementInfoList.json, StudioPartDefinition2.txt and StudioColorDefinition.txt. Geometry: Studio LDraw library plus the attributed official primitive included under ldraw/. Architecture and roof research notes accompany the model.

LDraw SHA-256: 32214b3a3d9c40f5b61b482e68b676df3c46fab8749cff6a8278cc78593ffc6c